

Zirui Wang

Research Interests: Coding Agent Post-Training · Agent Data and Training-Evaluation Loops

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EDUCATION

- **Zhejiang University** Sep. 2022 – Jul. 2025
Theoretical and Computational Chemistry M.S.
 - GPA: 4.5/5.0
 - Advisor: [Prof. Linjun Wang](#)
 - Research: Scientific ML, GNNs, Electronic Structure Modeling, and Molecular Dynamics
- **Lanzhou University** Sep. 2018 – Jul. 2022
Chemistry (National Class of Foundational Theory) B.S.
 - GPA: 3.8/4.0, Ranking: 14/280 (**Top 5%**)

EXPERIENCE

- **Valhalla Technology (Startup)** Mar. 2026 – Present
Artificial Intelligence Department AI Algorithm Engineer
 - **Scientific Agent Integration – MolClaw:** Extended BioClaw with an ODesign integration layer that maps natural-language requests to schema-validated task specifications for protein, ligand, and nucleic-acid design.
 - **Reliable Execution:** Packaged the ODesign environment with Docker/Conda and added dependency and checkpoint validation, automatic GPU selection, and recursive skill deployment, with completion checks based on expected output artifacts.
 - **LLM Evaluation – EpiBench:** Led the construction of a 1,609-sample, five-task benchmark with controlled sampling to reduce shortcut artifacts; evaluated nine LLMs and characterized failures in antibody-specific sequence grounding and residue localization.
- **Zhejiang University – Blockchain and Data Security Research Institute** Jul. 2025 – Mar. 2026
Department of Data Security for Large Language Models Algorithm Engineer
 - **Safety Data and SFT:** Constructed a 30K-prompt dataset covering safety-critical scenarios and fine-tuned Qwen2-7B-Instruct on 20K safety-alignment examples.
 - **MCTS-Guided Preference Optimization:** Applied DPO with MCTS-derived step-level reasoning signals and dual-reward optimization for safety alignment.
 - **Safety Evaluation:** Evaluated on StrongREJECT, with Goodness@0.1 increasing from 0.69 to 0.83.
- **ByteDance** Jan. 2025 – Mar. 2025
Monetization Product and Technology – Ads Algorithm Department Ads Algorithm Engineer Intern
 - **Style-Aware CVR Modeling:** Designed a dual-path architecture using MoE to encode ad-style features and cross-attention to model user-ad interactions.
 - **Gated Feature Fusion:** Applied Softmax gating to select among candidate style representations, then fused them with user profiles, contextual signals, and cross features for CVR prediction.
 - **Prediction Quality:** Reduced CVR prediction bias by 5.4%.

OPEN-SOURCE PROJECTS

- 🌐 [MolClaw](#), Scientific Agent System for Biomolecular Design, *Core Developer* [2026](#)
- 📊 [EpiBench](#), Epitope LLM Benchmark (**230 downloads**), *Lead / First Author* [2026](#)
- 🌐 [GNNs in Modern Drug Discovery](#), Open-Source Tutorial, *Chapter 1 Contributor* [2026](#)

PUBLICATIONS

Peer-Reviewed Publications

- [1] Wang, Zirui, Hou, Yusen and Liang, Shaofeng and Tian, Bowen and Zhang, Yanlin and Chen, Wenshuo and Yue, Yutao. **ABLE: Representing and Mapping LLMs via Attribution-Based Large-model Embedding**. *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026 Main Conference)*. **2026**.
[🌐 Webpage] [🌐 GitHub]
- Research highlight:** Led the development of ABLE, a training-free framework that aligns gradient-based feature attributions across tokenizers to construct comparable model embeddings. Evaluated 239 open-weight LLMs on model relationship prediction, routing, and benchmark score prediction, achieving Spearman correlations of 0.82–0.89 across five benchmarks in the latter task.

- [2] Niu, Kai and Wang, Chenyang and Zeng, Jiejun and **Wang, Zirui** and Liu, Yang and Wang, Linjun and Li, Cheng and Jin, Yizheng. **Ion Migration in Lead-Halide Perovskites: Cation Matters**. *The Journal of Physical Chemistry Letters*. 2024.
- [3] **Wang, Zirui** and Dong, Jiawei and Wang, Linjun. **Large-scale surface hopping simulation of charge transport in hexagonal molecular crystals: role of electronic coupling signs**. *Journal of Physics: Condensed Matter*. 2023.
- [4] Zhang, Hua and Wang, Zheyuan and **Wang, Zirui** and Chu, Yunpeng and Wang, Shuncheng and Hui, Xin-Ping. **Visible-light-mediated formal carbene insertion reaction: enantioselective synthesis of 1,4-dicarbonyl compounds containing an all-carbon quaternary stereocenter**. *ACS Catalysis*. 2022.

Preprints

- [5] **Wang, Zirui*** and Wang, Jiaqi* and Wang, Qinghan and Xu, Yuzhi and Du, Gang and Hou, Tingjun and Zhang, Odin. **EpiBench: Can LLMs Understand Epitopes for Antibody Drug Discovery?** *arXiv*. 2026.
[\[🌐 Webpage\]](#) [\[🔗 GitHub\]](#)

Research highlight: Led the design of EpiBench, a 1,609-sample benchmark spanning five epitope-centered antibody discovery tasks, with controlled sampling to reduce shortcut artifacts. Evaluated nine general-purpose LLMs under zero-shot protocols, revealing limitations in antibody-specific sequence grounding and inconsistent benefits from explicit reasoning across models and tasks.

CONFERENCE PRESENTATIONS

08, 2024 **The 15th National Conference of Theoretical and Computational Chemistry, Jilin.**

Universal Graph Neural Networks for the Hamiltonians of Perovskites. (Poster)

06, 2023 **The 33rd Chinese Chemistry Society Congress, Qingdao.**

Influence of Electronic Coupling Sign on Charge Transport in Hexagonal Lattices. (Poster)

SCHOLARSHIPS AND AWARDS

08, 2024 Outstanding Graduate Student, *Zhejiang University*.

12, 2023 Five-Star Graduate Student, *Zhejiang University*.

08, 2021 Hui-Chun Chin and Tsung-Dao Lee Chinese Undergraduate Research Endowment, *National*.

08, 2019 Provincial Third Prize in the National College Students' Mathematics Competition, *Gansu Province*.